

Chapter 8 — Long-Lived Assets

Homework Solutions

8.1 · Acquisition Cost of Equipment

Problem

Cedar Fair purchased two new rides for **\$26,000,000**, paying **\$13,000,000** cash and signing a note payable with 9% interest for the remaining amount. The note is payable in three years. Cedar Fair paid **\$125,000** for transportation and **\$1,200** freight insurance. In addition, the company paid **\$625,000** for installation of the ride.

How much of these expenditures related to the new rides purchase should be capitalized?

Equipment value to be capitalized

Item	Amount (\$)
Two rides	26,000,000
Transportation cost	125,000
Freight insurance	1,200
Installation cost	625,000
Total capitalized cost	26,751,200

Journal entry

Account	Debit	Credit
Ride equipment (+A)	26,751,200	
Cash (-A)		13,751,200
Note payable (+L)		13,000,000

8.2 · Capitalization of Interest During Construction

Problem

ABC Manufacturing is building a new factory. Construction began on January 1 and ended on December 31. During the year, ABC had the following:

- Construction expenditures: **\$5,000,000** (on average throughout the year)
- General debt: **\$10,000,000** at 6% interest
- Total interest incurred in the year: **\$800,000**

Question 1. What is the total asset related to the construction (i.e., the building)?

Step 1 — Calculate capitalizable interest

Apply the 6% interest rate to the average construction expenditures:

$$\text{\$5,000,000} \times 6\% = \text{\$300,000}$$

Step 2 — Compare to total interest incurred

Capitalizable interest: **\$300,000**

Total interest incurred: **\$800,000**

Capitalized interest must not exceed total interest incurred → **OK**

Journal entries during construction

Account	Debit	Credit
Construction in progress (+A)	5,000,000	
Cash (−A)		5,000,000
Construction in progress (+A)	300,000	
Cash (−A)		300,000

Disclosure

Item	Amount (\$)
Total interest incurred	800,000
Interest capitalized	300,000
Interest expensed	500,000
Construction in progress — ending balance	5,300,000

Question 2. Why do U.S. GAAP and IFRS require interest incurred during construction to be capitalized rather than expensed?

Reason	Explanation
1) Matching principle	Reflects cost in the same period as the revenue benefit
2) More accurate asset cost	Includes all costs necessary to get the asset ready for use

Reason	Explanation
3) Comparability	Ensures fair comparisons between firms

8.3 · Brand Names

Problem

Williams-Sonoma, Inc. is a specialty retailer of products for the home. The retail segment sells through its retail concepts: Williams-Sonoma, Pottery Barn, Pottery Barn Kids, PBteen, West Elm, and Rejuvenation. The direct-to-customer segment sells similar products through direct-mail catalogs and e-commerce websites.

The Williams-Sonoma brand name is well known in the United States. Discuss why the company does not report its brand name as an asset on its balance sheet. Is the brand name reflected anywhere in its financial statements?

Suggested solution

Intangible assets such as brand names are capitalized to the balance sheet **only when purchased** — that is, when there is a transaction with an independent party. Williams-Sonoma did not acquire its brand name in a purchase transaction; it built the brand with advertising, quality service and products, and word-of-mouth.

If Williams-Sonoma were subsequently acquired by another company, the brand would appear on the acquirer's balance sheet, because the brand name would have been part of the acquisition transaction. Internally generated brand names are therefore a form of **unreported (off-balance-sheet) asset**.

The brand name is not reflected anywhere on the financial statements. However, to the extent that it generates future cash flows, it **is** reflected in the market value of the company's share price even though it is not reported on the face of the balance sheet.

8.4 · R&D Failure and Share Prices

Problem

On August 5, 2016, *Bristol Myers Squibb* stock dropped by 17 percent after it announced that its drug failed in clinical trials as a treatment for lung cancer. The drug was already approved for treating some other types of cancer, with sales for those uses of nearly **\$1 billion**. Bristol was hoping to grow the market by getting approval for additional types of cancer. The drop reduced the company's market capitalization by **\$21 billion**.

Discuss the impact on the company's financial statements.

Suggested solution

Because research and development costs are **expensed when incurred**, the failure of the clinical trials had **no immediate impact** on Bristol Myers' financial statements — there were no capitalized R&D costs to write off following the announcement.

The decline in the share price reflected the reduction in future revenues, earnings, and cash flow that the capital market had already impounded into the share price in anticipation of successful trials. The price decline reflects the **forward-looking approach** the capital market uses to value shares: a company's share price today is a function of its future expected earnings and cash flow.